Asset Pricing Theory: S04 Production-Based Asset Pricing

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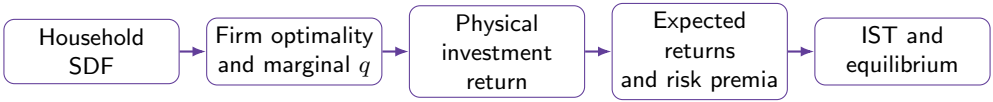
Structure of the course

- Email: amedeo.andriollo@dauphine.psl.eu
- Grading follows a 30-70 rule: 70% final exam, 30% project/homework (Nov. 6th).
- Office hours: feel "free" to (DO) come: B612.
- Slides/materials are updated regularly. Report a typo/mistake: corrections help the class.
- This session:
  - Firm optimality, marginal q, investment returns, expected returns, investment-specific technology, and equilibrium..
  - Main references: Campbell (2018, Ch.7); Cochrane (1991); Koijen and Van Nieuwerburgh (2020); Cochrane (2005, Ch.2.2 and 21.3).

$\trianglerighteq$ Market frictions.

Today's Route: From Investors to Firms

- **Sessions 1-2:** NA gives us a pricing restriction, while household/investor links the SDF to marginal utility. **Session 3:** Investors choose consumption and portfolios dynamically, given investment.
- **This session:** *What returns does physical investment generate, and how are those returns priced?*



- Household: marginal rate of substitution determines how future payoffs are valued across dates/states.

↪ **Firms:** marginal rate of transformation determines how future payoffs can be generated by investing.

▷ Firm optimality gives us a return on physical investment that must satisfy the SDF pricing restriction.

What Can Firm Optimality Tell Us?

- A firm decide: i) distribute resources to its shareholders today or
  ii) invest them in productive capital to generate future payoffs.
  ↪ **Trade-off:** investment reduces current distributions but increases future cash flows and value.
- At the optimum, (marginal) cost of investing must equal the present value of its (marginal) benefit:

$$\underbrace{\text{Marginal cost at } t}_{\text{foregone current payout}} = \mathbb{E}_t \left[M_{t+1} \underbrace{\text{Marginal payoff at } t+1}_{\text{cash flow + continuation value}} \right].$$

- Dividing by the current marginal cost gives us the familiar:

$$\boxed{1 = \mathbb{E}_t \left[M_{t+1} R^{I,g}_{t+1} \right],}$$

  where $R^{I,g}_{t+1}$ is the gross return on physical investment.

↪ Firm optimality allows us to construct an economically meaningful return that must be priced by the same SDF as financial assets, i.e. a **production-based** pricing restriction.

Back to Discrete Time

- Discrete time, $t = 0, 1, 2, \ldots$. The consumption good is our numeraire: all prices/returns are in real terms.
- **Agent decisions:** At date t, firms inherit the capital K_t and observe productivity Z_t. Given the info $\mathcal{F}_t$:
  - Firms choose labor L_t and investment I_t, thus determining current dividends D_t.
  - Households choose consumption and their portfolio holdings.

↪ Production at t uses the capital stock K_t and labor is chosen given the level of labor productivity Z_t. Investment undertaken today determines next period's K_{t+1}.

$$\underbrace{K_t, Z_t}_{\text{observed at } t} \longrightarrow \underbrace{L_t, I_t, D_t, K_{t+1}}_{\text{decisions and outcomes at } t} \longrightarrow \underbrace{Z_{t+1}}_{\text{new productivity shock}}$$

- As in the previous sessions, we distinguish gross and net returns:

$$R^g_{i,t+1} = \frac{P_{i,t+1} + D_{i,t+1}}{P_{i,t}}, \qquad R^{net}_{i,t+1} = R^g_{i,t+1} - 1,$$

where $P_{i,t}$ is the ex-dividend price and $D_{i,t+1}$ is the next-period payout.
↪ We will denote the gross return on physical investment by $R^{I,g}_{t+1}$.

Production, Investment, Dividends

- Firm combines capital K_t and effective labor $Z_t L_t$ to produce goods: $Y_t = F(K_t, Z_t L_t)$.
  The production function is such: $F_K > 0$, $F_L > 0$, $F_{KK} < 0$, $F_{LL} < 0$. Real wage is w_t.
- Investment increases the capital next period using resources that could be distributed to shareholders:

$$K_{t+1} = (1-\delta)K_t + I_t, \qquad D_t = Y_t - w_t L_t - I_t - G(I_t, K_t),$$

  where $\delta \in (0,1)$ is the depreciation rate and $G(I_t, K_t)$ captures the **adjusting** investment costs:

$$G(I,K) = \frac{\kappa}{2}\left(\frac{I}{K} - \delta\right)^2 K, \qquad \kappa > 0, \quad K > 0.$$

  - Adjustment costs are zero when investment exactly replaces depreciated capital, $I/K = \delta$.
  - Deviations from this replacement rate are costly. A larger κ implies greater adjustment costs.

! Adjustment costs reduce current dividends w/out creating capital tomorrow.

▷ We initially assume interior investment choices with positive marginal investment costs.

The Household Side (1/2)

- The household derives utility from consumption C_t and leisure $\ell_t = 1 - L_t^s$ (L_t^s is labor supply):

$$\max_{\{C_t, L_t^s, \theta_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t, \ell_t), \qquad 0 < \beta < 1, \quad u_C, u_\ell > 0.$$

- At each date, the household allocates its resources between consumption and financial assets:

$$\underbrace{C_t + \sum_i P_{i,t}\theta_{i,t}}_{\text{consumption and portfolio purchases}} = \underbrace{w_t L_t^s + \sum_i (P_{i,t} + D_{i,t})\theta_{i,t-1}}_{\text{labor income and existing portfolio value}},$$

  where $P_{i,t}$ is the ex-dividend price, $D_{i,t}$ the dividend, and $\theta_{i,t}$ the holdings chosen at t.

- Let B_t denote the difference between the RHS and LHS of the budget constraint. The Lagrangian is:

$$\mathcal{L}^H = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[u(C_t, 1 - L_t^s) + \eta_t B_t \right],$$

- At an interior optimum, the first-order conditions imply:

$$\frac{\partial L^H}{\partial C_t} = 0 \iff u_C(C_t, \ell_t) = -\eta_t, \quad \frac{\partial L^H}{\partial l_t} = 0 \iff u_l(C_t, \ell_t) = -\eta_t w_t, \quad \boxed{\frac{u_\ell(C_t, \ell_t)}{u_C(C_t, \ell_t)} = w_t.}$$

The Household Side (2/2): SDF

- The household decides how many shares $\theta_{i,t}$ of asset i to hold:
  - Purchasing one additional share costs $P_{i,t}$ units of consumption today.
  - Next period, the share delivers $P_{i,t+1} + D_{i,t+1}$, whose value is uncertain at t.
- FOC wrt $\theta_{i,t}$, and by virtue of law of iterated expectations:

$$\underbrace{\eta_t P_{i,t}}_{\text{marginal utility cost today}} = \beta \mathbb{E}_t \left[\underbrace{\eta_{t+1}(P_{i,t+1} + D_{i,t+1})}_{\text{marginal utility benefit tomorrow}} \right].$$

- Recall that at the optimum: $u_C(C_t, \ell_t) = -\eta_t$. Dividing this first-order condition by $\eta_t P_{i,t}$ gives:

$$1 = \mathbb{E}_t \left[M_{t+1} R^g_{i,t+1} \right], \qquad M_{t+1} = \beta \frac{u_C(C_{t+1}, \ell_{t+1})}{u_C(C_t, \ell_t)}.$$

The Firm Side (1/4)

- A firm acts in the interest of its shareholders by choosing labor and investment to maximize its profit stream: the present value of its future dividends, taking the household's SDF M_{t+1} as given.
- Notation: $\Lambda_0 = 1$, $\Lambda_t = \prod_{s=1}^{t} M_s$, $\Lambda_{t+j}/\Lambda_t := M_{t,t+j}$, discounting payoffs at $t+j$ back to date t. Let V_t denote the **value of the firm**:

$$V_t(K_t) = \max_{\{I_{t+j}, L_{t+j}\}_{j\geq 0}} \mathbb{E}_t \left[\sum_{j=0}^{\infty} M_{t,t+j} D_{t+j} \right] = \max_{I_t, L_t} \left\{ D_t + \mathbb{E}_t \left[M_{t+1} V_{t+1}(K_{t+1}) \right] \right\}.$$

- Post-dividend value: firm's **mkt price** after the current dividend has been distributed is

$$P_t = V_t - D_t = \mathbb{E}_t \left[M_{t+1} V_{t+1} \right].$$

↪ The firm chooses investment by comparing the resources distributed to shareholders today with the SDF-discounted value of the additional future dividends that investment generates.

- subject to the constraints:

$$D_t = F(K_t, Z_t L_t) - w_t L_t - I_t - G(I_t, K_t),$$
$$K_{t+1} = (1-\delta) K_t + I_t.$$

The Firm Side (2/4): q

- The Lagrangian introduces a multiplier q_t on the capital accumulation constraint:

$$\mathcal{L}^F = \mathbb{E}_0 \sum_{t=0}^{\infty} \Lambda_t \left[D_t + q_t \big((1-\delta) K_t + I_t - K_{t+1} \big) \right].$$

↪ Economic interpretation of q_t:

  - Each period's own capital accumulation constraint (time-varying and state-dependent).
  - It is the **shadow value of installed capital**: the value at t of an additional unit of productive capital available at $t+1$.
  - Multiplying q_t by Λ_t expresses this value in date-0 value: (q_t is measured in date-t consumption units).

- Optimal investment level:

$$\frac{\partial \mathcal{L}^F}{\partial I_t} = 0 \Longrightarrow -1 - G_I(I_t, K_t) + q_t = 0 \Longrightarrow \boxed{q_t = 1 + G_I(I_t, K_t).}$$

  Because $G_I(I_t, K_t) = \kappa \left(I_t / K_t - \delta \right)$, then $I_t / K_t = \delta + (q_t - 1)/\kappa$.
  Higher marginal q encourages investment; larger adjustment costs (higher κ) reduce its responsiveness.

- Optimal labor demand:

$$\frac{\partial \mathcal{L}^F}{\partial L_t} = 0 \Longrightarrow \boxed{Z_t F_2(K_t, Z_t L_t) = w_t.}$$

The Firm Side (3/4): Value of Inherited Capital

- *What is the marginal value of an additional unit of capital inherited at $t+1$?*
  At $t+1$, capital K_{t+1} is predetermined, while labor L_{t+1} and investment I_{t+1} are optimally chosen.
- Recall $\mathcal{L}^F$, its date-$(t+1)$ contribution is:

$$\Lambda_{t+1}\left[D_{t+1} + q_{t+1}\big((1-\delta)K_{t+1} + I_{t+1} - K_{t+2}\big)\right].$$

  An additional unit of K_{t+1} affects both dividends and future capital.

  - **current dividends:** $\dfrac{\partial D_{t+1}}{\partial K_{t+1}} = \underbrace{F_{K,t+1}}_{\text{additional output}} - \underbrace{G_{K,t+1}}_{\text{effect on adjustment costs}}$.
  - **Future capital:** $K_{t+2} = (1-\delta)K_{t+1} + I_{t+1} \implies \frac{\partial K_{t+2}}{\partial K_{t+1}} = 1-\delta.$
    Thus, an additional unit of inherited capital leaves $(1-\delta)$ units available at $t+2$, valued at q_{t+1} per unit.
- **The envelope condition:** Differentiating the Lagrangian contribution wrt inherited capital, and dividing by Λ_{t+1}:

$$V_{K,t+1} = \frac{\partial D_{t+1}}{\partial K_{t+1}} + q_{t+1}(1-\delta) = \underbrace{F_{K,t+1} - G_{K,t+1}}_{\text{marginal dividends}} + \underbrace{(1-\delta)q_{t+1}}_{\text{continuation value.}}$$

  By the envelope theorem, we evaluate the derivative at optimal choices.

▷ $V_{K,t+1}$ is the marginal payoff of capital at $t+1$.

The Firm Side (4/4): The Capital Euler Eq.

- From the marginal payoff $V_{K,t+1}$ to the **intertemporal Lagrangian** $\mathcal{L}^F$ to determine its value at t.
- **Capital FOC:** K_{t+1} enters in two dates:
  1. At t, it affects negatively, contributing $-\Lambda_t q_t$.
  2. At $t+1$, it affects dividends and future capital accumulation, contributing $\Lambda_{t+1} V_{K,t+1}$.

↪ Differentiating the $\mathcal{L}^F$ wrt K_{t+1} gives:

$$\underbrace{-\Lambda_t q_t}_{\text{marginal cost at } t} + \underbrace{\Lambda_{t+1}\left[F_{K,t+1} - G_{K,t+1} + (1-\delta)q_{t+1}\right]}_{\text{marginal benefit at } t+1}.$$

- Since K_{t+1} is chosen using information available at t:

$$0 = \mathbb{E}_t\left[-\Lambda_t q_t + \Lambda_{t+1} V_{K,t+1}\right] = \Lambda_t\left\{-q_t + \mathbb{E}_t[M_{t+1} V_{K,t+1}]\right\},$$

- Dividing by $\Lambda_t > 0$, and substituting the envelope condition, yields the **capital Euler equation**:

$$\boxed{q_t = \mathbb{E}_t[M_{t+1} V_{K,t+1}] = \mathbb{E}_t\left[M_{t+1}\left(F_{K,t+1} - G_{K,t+1} + (1-\delta)q_{t+1}\right)\right].}$$

↪ The marginal cost q_t of installing capital today equals the SDF-discounted marginal payoff $V_{K,t+1}$ tomorrow.

Return on Investment (1/2)

- Following Cochrane (1991), consider a small increase in investment at t: $\mathrm{d}I_t = \varepsilon > 0$.
- **At date t:** Higher investment raises next-period capital but reduces current dividends:

$$\mathrm{d}K_{t+1} = \varepsilon, \qquad \mathrm{d}D_t = -(1 + G_{I,t})\varepsilon = -q_t\varepsilon.$$

- **At date $t+1$:** The additional capital generates more output. However, we want to isolate the payoff over one period, w/out changing the capital stock from $t+2$ onward.
  $\hookrightarrow$ The firm can reduce investment at $t+1$, using the additional capital accumulated at t:

$$\begin{aligned} K_{t+2} &= (1-\delta)K_{t+1} + I_{t+1}, \\ \mathrm{d}K_{t+2} &= (1-\delta)\underbrace{\mathrm{d}K_{t+1}}_{\varepsilon} + \mathrm{d}I_{t+1} = 0 \quad \Longrightarrow \quad \boxed{\mathrm{d}I_{t+1} = -(1-\delta)\varepsilon}. \end{aligned}$$

- **Effect on dividends at $t+1$:** The additional capital increases production, changes adjustment costs, and allows the firm to save on investment expenditure:

$$\begin{aligned} \mathrm{d}D_{t+1} &= \underbrace{(F_{K,t+1} - G_{K,t+1})\varepsilon}_{\text{marginal dividends from capital}} - \underbrace{(1 + G_{I,t+1})\mathrm{d}I_{t+1}}_{\text{saving on investment}} \\ &= \left[F_{K,t+1} - G_{K,t+1} + (1-\delta)q_{t+1}\right]\varepsilon. \end{aligned}$$

Return on Investment (2/2)

- Gross investment return: the ratio of the marginal payoff at $t+1$ to the marginal cost at t gives:

$$R^{I,g}_{t+1} = \frac{\mathrm{d}D_{t+1}}{-\mathrm{d}D_t} = \frac{\overbrace{F_{K,t+1} - G_{K,t+1}}^{\text{marginal cash flow}} + \overbrace{(1-\delta)q_{t+1}}^{\text{continuation value}}}{\underbrace{q_t}_{\text{marginal cost at } t}}.$$

↪ **intertemporal marginal rate of transformation**.
Notice that the continuation value $(1-\delta)q_{t+1}$ comes from the reduction in future investment.

- The pricing equation: recall the capital Euler equation derived from firm optimality:

$$q_t = \mathbb{E}_t\left[M_{t+1}\left(F_{K,t+1} - G_{K,t+1} + (1-\delta)q_{t+1}\right)\right].$$

Dividing by $q_t > 0$ gives: $1 = \mathbb{E}_t\left[M_{t+1}\frac{V_{K,t+1}}{q_t}\right] \implies 1 = \mathbb{E}_t[M_{t+1}R^{I,g}_{t+1}]$.

- **W/out adjustment costs:** If $G(I,K) = 0$, then $q_t = q_{t+1} = 1$ (FOC investment), implying:

$$R^{I,g}_{t+1} = F_{K,t+1} + 1 - \delta.$$

! This is a marginal investment return, not necessarily the firm's stock return.

From Physical Investment to Financial Returns

- The firm does not pin down the SDF:

$$1 = \mathbb{E}_t[M_{t+1} R^{I,g}_{t+1}] \quad \not\Longrightarrow \quad M_{t+1} = \frac{1}{R^{I,g}_{t+1}}.$$

To see why, suppose there are two possible states at $t+1$, with probabilities π_1 and π_2:

$$\pi_1 M_1 R^{I,g}_1 + \pi_2 M_2 R^{I,g}_2 = 1.$$

$\hookrightarrow$ one equation for two unknown states for M.

  - A marginal investment decision transfers resources across dates, but not generally across states.

- Can a portfolio replicate physical investment? Suppose a traded portfolio delivers exactly the payoff $R^{I,g}_{t+1}$ in every state, then:

$$p^{\text{mimic}}_t = \mathbb{E}_t[M_{t+1} R^{I,g}_{t+1}] = 1 \quad \Longrightarrow \quad \boxed{R^{\text{mimic},g}_{t+1} = \frac{R^{I,g}_{t+1}}{p^{\text{mimic}}_t} = R^{I,g}_{t+1}.}$$

Complete markets guarantee that such a portfolio exists, but spanning this particular payoff is **sufficient**.

$\hookrightarrow$ **The distinction:** Exact payoff replication implies equality of physical investment and portfolio returns **state by state**. Being priced by the same SDF does not, by itself, imply this equality.

$\trianglerighteq$ When firm's **own stock return** = physical investment return? Homogeneity properties of production.

In-Class Exercise 1

- Consider a competitive firm with an initial capital stock $K_t = 100$, a depreciation rate $\delta = 0.10$, and:

$$G(I, K) = \frac{\kappa}{2}\left(\frac{I}{K} - 0.10\right)^2 K, \qquad \kappa = 4.$$

- Suppose next-period marginal conditions are known at date t: (no uncertainty between t and $t+1$)

$$\frac{I_{t+1}}{K_{t+1}} = 0.20, \qquad F_{K,t+1} = 0.18, \qquad R^g_{f,t} = 1.25.$$

- **Questions:**
  1. **Marginal adjustment costs:** Derive $G_I(I, K)$ and $G_K(I, K)$.
  2. **Marginal value of capital:** Using the firm's optimality conditions, compute q_{t+1} and the marginal payoff $V_{K,t+1}$.
  3. **Optimal investment and returns:** Determine q_t and the optimal investment I_t. Compute the gross and net returns on investment, verifying that the capital Euler equation holds.
  4. **An incorrect investment decision:** Suppose the manager ignores the adjustment-cost term $-G_{K,t+1}$ when evaluating the marginal payoff. How does this affect the estimated q_t and the resulting investment decision?

Exercise 1 Solution (1/2)

- **Marginal adjustment costs:** Recall that $G(I,K) = \frac{\kappa}{2}K(x-\delta)^2$.
  If we set $x = \frac{I}{K}$, then: $\frac{\partial x}{\partial I} = \frac{1}{K}$ and $\frac{\partial x}{\partial K} = -\frac{I}{K^2} = -\frac{x}{K}$.
  - **Derivative wrt investment:** Holding K fixed: $G_I(I,K) = \frac{\kappa}{2}K \cdot 2(x-\delta)\frac{1}{K} = \kappa(x-\delta)$.
  - **Derivative wrt capital:** Holding investment I fixed, both K and the investment rate $x = I/K$ change:

$$G_K(I,K) = \frac{\kappa}{2}(x-\delta)^2 + \frac{\kappa}{2}K \cdot 2(x-\delta)\left(-\frac{x}{K}\right) = \frac{\kappa}{2}(\delta^2 - x^2).$$

- **Marginal value of capital at $t+1$:** Using $x_{t+1} = 0.20$ and $\kappa = 4$, the investment FOC gives

$$q_{t+1} = 1 + 4(0.20 - 0.10) = 1.40, \qquad G_{K,t+1} = 2(0.10^2 - 0.20^2) = -0.06.$$

  Applying the envelope condition:

$$\begin{aligned} V_{K,t+1} &= F_{K,t+1} - G_{K,t+1} + (1-\delta)q_{t+1} \\ &= \underbrace{0.18 + 0.06}_{\text{marginal cash flow } = 0.24} + \underbrace{0.90(1.40)}_{\text{continuation value } = 1.26} = \boxed{1.50}. \end{aligned}$$

↪ **Economic interpretation:** Holding investment I_{t+1} fixed, more capital reduces the investment rate I_{t+1}/K_{t+1}, bringing it closer to the costless replacement rate $\delta = 0.10$. This reduction in adjustment intensity outweighs the direct effect of increasing the scale of capital:

$$G_{K,t+1} = \underbrace{0.02}_{\text{scale effect}} - \underbrace{0.08}_{\text{lower investment intensity}} = -0.06.$$

Thus, an additional unit of capital saves 0.06 in adjustment costs, on top of its contribution to output.

Exercise 1 Solution (2/2)

- **The marginal value of investment at t:** With no uncertainty, the SDF is given by the inverse of the gross risk-free return:
$$M_{t+1} = \frac{1}{R^g_{f,t}} = \frac{1}{1.25} = 0.80.$$
Using the capital Euler equation and $V_{K,t+1} = 1.50$: $q_t = \mathbb{E}_t[M_{t+1} V_{K,t+1}] = 1.20$.
- **Optimal investment:** From the investment FOC, $q_t = 1 + G_I(I_t, K_t)$:
$$\frac{I_t}{K_t} = \delta + \frac{q_t - 1}{\kappa} = 0.10 + \frac{1.20 - 1}{4} = 0.15 \implies I_t = 100(0.15) = 15.$$
- **The return on physical investment:** The ratio of the marginal payoff tomorrow to the marginal cost today gives: $R^{I,g}_{t+1} = V_{K,t+1}/q_t = 1.50/1.20 = 1.25$. The net return is 25%, and the Euler equation is verified: $\mathbb{E}_t[M_{t+1} R^{I,g}_{t+1}] = 0.80(1.25) = 1$.
- **What if the manager ignores $-G_{K,t+1}$?** Omitting the adjustment-cost benefit understates the payoff:
$$\breve{V}_{K,t+1} = 0.18 + 0.90(1.40) = 1.44, \quad \breve{q}_t = 0.80(1.44) = 1.152, \quad \breve{I}_t = 100\left(0.10 + \frac{1.152 - 1}{4}\right) = 13.8.$$

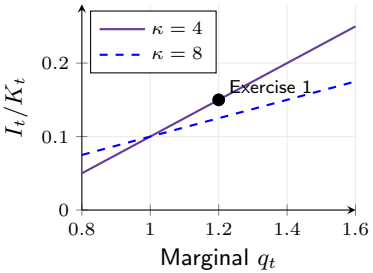
↪ The manager underinvests by $15 - 13.8 = 1.2$ units. Additional investment would increase firm value.

q-Theory: Value and Investment

- From the firm's FOC, the marginal value of capital equals its marginal cost: $\frac{I_t}{K_t} = \delta + \frac{q_t - 1}{\kappa}$.
- **Investment responsiveness:** Holding technology fixed:

$$\frac{\partial (I_t/K_t)}{\partial q_t} = \frac{1}{\kappa} > 0.$$

- Recall: $q_t = \mathbb{E}_t[M_{t+1} V_{K,t+1}]$.
  Marginal q reflects both i) future marginal payoffs and ii) how households discount them across states. Changes in i) profitability or ii) discount rates can affect investment today.



↪ **What should we take home from the exercise?**

1. Investment depends on marginal valuation and adjustment costs. With $q_t = 1.20$, the investment rate is 15% for $\kappa = 4$, but only 12.5% for $\kappa = 8$. The same valuation induces less investment when capital adjustment is more costly.
2. The entire marginal payoff matters, Omitting the adjustment-cost saving $-G_{K,t+1}$ understates the marginal value of capital ($q_t = 1.152$ instead of 1.20), reducing investment from 15 to 13.8.

▷ **The central insight of q-theory:** the firm adjusts its capital stock until the marginal cost of investing today equals the discounted marginal benefit of future capital.

Marginal q VS. Average Q

- Investment depends on marginal q_t: the shadow value of an additional unit of installed capital.
  Can we relate this unobservable marginal value to the firm's market value?
- **Marginal vs. average:** q_t measures the value of an additional unit of capital installed for $t+1$, $Q_t = P_t/K_{t+1}$ measures the firm's mkt value per unit of (installed) capital.
- **Constant returns to scale:** Suppose production and adjustment costs are homogeneous of degree one:
  - Scaling capital, labor, and investment by $a > 0$ scales dividends and firm value proportionally, $V_t(aK_t) = aV_t(K_t)$.
  - By Euler's theorem for homogeneous functions:

$$V_t(K_t) = V_{K,t}K_t \quad \Longrightarrow \quad V_{K,t} = \frac{V_t}{K_t}.$$

- **From marginal q to average Q:** Using the capital Euler equation and the homogeneity result:

$$q_t = \mathbb{E}_t[M_{t+1}V_{K,t+1}] = \mathbb{E}_t\left[M_{t+1}\frac{V_{t+1}}{K_{t+1}}\right] = \frac{\mathbb{E}_t[M_{t+1}V_{t+1}]}{K_{t+1}} = \frac{P_t}{K_{t+1}} \equiv Q_t,$$

  where K_{t+1} is known at t and P_t is the firm's post-dividend mkt value.

↪ Under homogeneity, **marginal and average valuations coincide**.
Notice the timing: $V_{K,t}$ values inherited capital, $q_t = Q_t = P_t/K_{t+1}$ values capital installed for next period.

▷ Without homogeneity, fixed costs,.. marginal $q \neq$ average Q.
(Empirical market-to-book ratios may also differ Q_t).

When Do Investment and Stock Returns Coincide?

- Under constant returns to scale (CRS), marginal q equals average Q:

$$\underbrace{q_t = P_t/K_{t+1}}_{\text{marginal } q=\text{average } Q}, \qquad \underbrace{V_{K,t+1} = V_{t+1}/K_{t+1}}_{\text{marginal value=average value}}.$$

- *Does this equality imply that the firm's stock return coincides with its physical investment return?*
  Consider the same homogeneous, single-capital, all-equity firm:
  - The **stock return** compares the firm's value at $t+1$, including dividends, with its post-dividend market value at t.
  - The **investment return** compares the marginal payoff of installed capital at $t+1$ with its marginal cost at t.
- Using $V_{t+1} = P_{t+1} + D_{t+1}$ and the homogeneity results:

$$R^{S,g}_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t} = \frac{V_{t+1}}{P_t} = \underbrace{\frac{V_{t+1}/K_{t+1}}{P_t/K_{t+1}}}_{\text{ratio of average valuations}} \overset{\text{CRS}}{=} \underbrace{\frac{V_{K,t+1}}{q_t}}_{\text{ratio of marginal valuations}} = R^{I,g}_{t+1}.$$

↪ Under homogeneity, the average valuations determining the stock return coincide with the marginal valuations determining the investment return, thus $R^{S,g}_{t+1} = R^{I,g}_{t+1}$ **state by state**.

Production-Based Risk Premia

- Recall: $1 = \mathbb{E}_t[M_{t+1} R^{I,g}_{t+1}]$. Using the definition of conditional covariance:

$$1 = \mathbb{E}_t[M_{t+1}]\mathbb{E}_t[R^{I,g}_{t+1}] + \mathrm{Cov}_t(M_{t+1}, R^{I,g}_{t+1}).$$

  Since $R^g_{f,t} = 1/\mathbb{E}_t[M_{t+1}]$,

$$\mathbb{E}_t[R^{I,g}_{t+1}] - R^g_{f,t} = -\frac{\mathrm{Cov}_t(M_{t+1}, R^{I,g}_{t+1})}{\mathbb{E}_t[M_{t+1}]}.$$

  ↪ As for financial assets, expected excess returns depend on the covariance between returns and the SDF.

- **What does production add?** Recall that the return on physical investment is:

$$R^{I,g}_{t+1} = \frac{V_{K,t+1}}{q_t}, \qquad V_{K,t+1} = F_{K,t+1} - G_{K,t+1} + (1-\delta)q_{t+1}.$$

  Since q_t is known at t, we can express the risk premium in terms of the marginal payoff of capital:

$$\boxed{\mathbb{E}_t[R^{I,g}_{t+1}] - R^g_{f,t} = -\frac{\mathrm{Cov}_t(M_{t+1}, V_{K,t+1})}{q_t \mathbb{E}_t[M_{t+1}]}.}$$

↪ Physical investment commands a positive risk premium when its marginal payoff is low in states where M_{t+1} is high, i.e., when households value additional consumption the most.
↪ Investment provides little payoff when consumption is most valuable.

▷ Production determines the **marginal payoff** VS. the SDF determines the **risk premium** through comovements.

From Technology to Firm Characteristics

- Production: Consider a Cobb-Douglas technology with constant returns to scale:

$$Y_{i,t} = Z_{i,t} K_{i,t}^{\alpha} L_{i,t}^{1-\alpha}, \qquad F_{K,i,t} = \alpha Z_{i,t} K_{i,t}^{\alpha-1} L_{i,t}^{1-\alpha} = \alpha Y_{i,t}/K_{i,t}, \qquad 0 < \alpha < 1.$$

  $\hookrightarrow$ The marginal product of capital is proportional to the output-to-capital ratio, a measure of firm profitability.

- Let $x_{i,t} = I_{i,t}/K_{i,t}$. Recall: $q_{i,t} = 1 + \kappa(x_{i,t} - \delta), \quad -G_{K,i,t} = \frac{\kappa}{2}(x_{i,t}^2 - \delta^2)$.

- **The physical investment return:** Substituting into $R_{i,t+1}^{I,g} = [F_{K,i,t+1} - G_{K,i,t+1} + (1-\delta)q_{i,t+1}]/q_{i,t}$ gives:

$$R_{i,t+1}^{I,g} = \frac{\overbrace{\alpha Y_{i,t+1}/K_{i,t+1}}^{\text{marginal output}} + \overbrace{\frac{\kappa}{2}(x_{i,t+1}^2 - \delta^2)}^{\text{adjustment-cost effect}} + (1-\delta)[1 + \kappa(x_{i,t+1} - \delta)]}{\underbrace{1 + \kappa(x_{i,t} - \delta)}_{\text{marginal investment cost at } t}}.$$

$\hookrightarrow$ **How to read it:**
i) Current investment $x_{i,t}$ determines the marginal cost of installed capital ($q_{i,t}$).
ii) Future profitability $Y_{i,t+1}/K_{i,t+1}$ and investment $x_{i,t+1}$ determine its marginal payoff.

$\trianglerighteq$ Under homogeneity, this also characterizes the firm's stock return.

Investment, Profitability, and Expected Returns

- *How do investment and profitability relate to expected returns?*
  Recall that the investment return is the marginal payoff of capital relative to its marginal cost:

$$\mathbb{E}_t[R^{I,g}_{i,t+1}] = \mathbb{E}_t[V_{K,i,t+1}]/q_{i,t} = \frac{\mathbb{E}_t[V_{K,i,t+1}]}{1+\kappa(I_{i,t}/K_{i,t}-\delta)}$$

  - **Investment:** Holding the expected marginal payoff $\mathbb{E}_t[V_{K,i,t+1}] > 0$ fixed:

$$\frac{\partial \mathbb{E}_t[R^{I,g}_{i,t+1}]}{\partial x_{i,t}} = -\frac{\kappa \mathbb{E}_t[V_{K,i,t+1}]}{[1+\kappa(x_{i,t}-\delta)]^2} < 0.$$

    ↑ investment implies ↑ marginal cost $q_{i,t}$ and ↓ expected return.
  - **Profitability:** Holding current investment and the other components of the marginal payoff fixed:

$$\partial \mathbb{E}_t[R^{I,g}_{i,t+1}]/\partial \mathbb{E}_t[F_{K,i,t+1}] = 1/q_{i,t} > 0.$$

    ↑ expected marginal profitability implies ↑ expected returns.

- **Economic interpretation:**
  - For a given expected payoff, required ↓ return raises the value of installed capital (q_t ↑), encouraging investment (I_t/K_t ↑).
  - Conversely, ↑ expected profitability combined with ↓ investment implies ↑ expected return: the firm must earn a sufficiently high return to justify investing so little despite attractive future payoffs.

▷ Conditional comparative statics: profitability, investment, future payoffs, and risk may change together.

Pricing: Household Demand and Firm Supply

- *How do households' valuation of risk and firms' investment decisions jointly determine expected returns?*

Household

$$\mathbb{E}_t[R^{I,g}_{t+1}] - R^g_{f,t} = -\frac{\mathrm{Cov}_t(M_{t+1}, R^{I,g}_{t+1})}{\mathbb{E}_t[M_{t+1}]}.$$

- The SDF captures households' marginal valuation of consumption across dates and states.
- Expected returns compensate for exposure to risks.

Firm

$$\mathbb{E}_t[R^{I,g}_{t+1}] = \frac{\mathbb{E}_t[V_{K,t+1}]}{1+\kappa(x_t-\delta)}.$$

- Production determines the marginal payoff of capital through profitability and investment opportunities.
- Firm optimality links expected returns to investment through marginal q.

- **At equilibrium:**

$$\underbrace{1+\kappa(x_t-\delta)}_{\text{marginal investment cost } q_t} = \mathbb{E}_t[M_{t+1}V_{K,t+1}] = \underbrace{\mathbb{E}_t[M_{t+1}]\mathbb{E}_t[V_{K,t+1}]}_{\text{discounted expected payoff}} + \underbrace{\mathrm{Cov}_t(M_{t+1}, V_{K,t+1})}_{\text{risk adjustment}}.$$

↪ **Production-based asset pricing:** Firms' profitability and investment decisions restrict expected returns, while the SDF determines how the associated risk is priced.

▷ The **investment CAPM** links expected returns to investment and profitability.

From Production-Based Pricing to Factor Models

- **A linear projection of the SDF:** Let f_{t+1} denote a vector of investment-related factors (e.g., physical investment returns or innovations in investment opportunities):

$$M_{t+1} = a_t + b_t' f_{t+1} + \varepsilon_{t+1}, \qquad \mathbb{E}_t[\varepsilon_{t+1}] = 0, \quad \mathrm{Cov}_t(\varepsilon_{t+1}, f_{t+1}) = 0.$$

- Substituting into the SDF:

$$\mathbb{E}_t[R^g_{i,t+1}] - R^g_{f,t} = -\frac{b_t' \,\mathrm{Cov}_t(f_{t+1}, R^g_{i,t+1}) + \mathrm{Cov}_t(\varepsilon_{t+1}, R^g_{i,t+1})}{\mathbb{E}_t[M_{t+1}]}.$$

  $\hookrightarrow$ Orthogonality to f_{t+1} alone is not sufficient to claim pricing:
  the projection residual must be uncorrelated with the returns of all assets being priced.

- Suppose an investment-return factor: $M_{t+1} = a_t - b_t R^{I,g}_{t+1} + \varepsilon_{t+1}$.
  If the residual is unpriced, and using the investment Euler equation:

$$\mathbb{E}_t[R^g_{i,t+1}] - R^g_{f,t} = \frac{b_t}{\mathbb{E}_t[M_{t+1}]} \mathrm{Cov}_t(R^{I,g}_{t+1}, R^g_{i,t+1}) = \underbrace{\frac{\mathrm{Cov}_t(R^g_{i,t+1}, R^{I,g}_{t+1})}{\mathrm{Var}_t(R^{I,g}_{t+1})}}_{\beta^I_{i,t}} \underbrace{\left(\mathbb{E}_t[R^{I,g}_{t+1}] - R^g_{f,t}\right)}_{\text{investment risk premium}}.$$

  Greater exposure to the investment return earn higher expected returns if positive premium.

! **Current q_t and I_t/K_t are known at t**: they may predict expected returns, but have zero conditional covariance with future returns (future investment returns or innovations can serve as risk factors).

▷ Firm optimality motivates the factor, but does not guarantee that the residual is unpriced.

Few Empirical Evidence

- **Aggregate time series: Cochrane (1991).** Investment returns from aggregate investment data and an adjustment-cost technology; comparison with real stock returns (1947Q1–1987Q4):

$$\mathrm{Corr}(R^{I,g}, R^{S,g}) = \begin{cases} 0.241 & \text{quarterly returns,} \\ 0.385 & \text{overlapping annual returns.} \end{cases}$$

  - Investment and stock returns exhibit significant comovement and similar predictability.
  - ! corr is not one; plus dividend-price ratio also predicts a stock-return component not captured by investment returns.

- **Structural cross-sectional tests: Liu, Whited and Zhang (2009).**
  Infer investment returns from accounting data and estimated technology parameters.
  - Average returns across several characteristic-sorted portfolios. But matching expected returns and return variances is more demanding: expected-return errors remain substantial.

- **Empirical factor models: Hou, Xue and Zhang (2015).**
  The q-**factor model** uses traded portfolios based on market, size, investment, and profitability.
  - It does not directly test the equality of physical investment and stock returns.

↪ Production provides economically motivated restrictions. However, comovement, matching return moments, and factor-pricing success are **different tests**.
None establishes the exact state-by-state equality $R^{S,g} = R^{I,g}$ or uniquely identifies the SDF.

Investment-Specific Technology (IST)

- So far, one unit of investment expenditure produces one unit of installed capital.
  What if technological progress changes the efficiency of investment?
- Let J_t denote investment expenditure in consumption units and $\mu_t > 0$ the amount of capital installed per unit of expenditure:

$$K_{t+1} = (1-\delta)K_t + \mu_t J_t, \qquad D_t = Y_t - w_t L_t - J_t - \widetilde{G}(J_t, K_t).$$

  $\hookrightarrow$ An increase in μ_t allows the firm to install more capital for a given expenditure.

- We maintain the quadratic resource-cost specification: $\widetilde{G}(J,K) = \frac{\kappa}{2}\left(\frac{J}{K} - \delta\right)^2 K$.
  $\hookrightarrow$ Adjustment costs depend on **investment expenditure** J, rather than μJ.
- **An equivalent interpretation.** Define investment $I_t^p = \mu_t J_t$ and its relative price $p_t^I = 1/\mu_t$, measured in consumption units. Then:

$$K_{t+1} = (1-\delta)K_t + I_t^p, \qquad D_t = Y_t - w_t L_t - p_t^I I_t^p - \widetilde{G}(p_t^I I_t^p, K_t).$$

  $\hookrightarrow$ Higher investment efficiency ($\mu_t \uparrow$) is equivalent to a lower relative price of installed capital ($p_t^I \downarrow$).

$\trianglerighteq$ μ_t is observed when investment is chosen at t, while μ_{t+1} is uncertain.
IST shocks therefore affect both investment decisions and future marginal payoffs.

IST (1/2): Investment and Marginal q

- J_t expenditure VS. $\mu_t J_t$ amount.
- With IST, the date-t contribution to $\mathcal{L}^F$ becomes:

$$\Lambda_t \left[D_t + q_t \big((1-\delta) K_t + \mu_t J_t - K_{t+1} \big) \right].$$

  q_t is the shadow value of one additional unit of capital installed for $t+1$, in consumption units.
- Investment FOC: one additional unit of expenditure costs $1 + \widetilde{G}_{J,t}$, but installs μ_t units of capital:

$$\frac{\partial \mathcal{L}^F}{\partial J_t} = 0 \Longrightarrow -1 - \widetilde{G}_{J,t} + \mu_t q_t = 0 \Longrightarrow q_t = \frac{1 + \widetilde{G}_{J,t}}{\mu_t}.$$

  $\hookrightarrow$ Marginal q equals the marginal expenditure cost per unit of installed capital.
- Let $j_t = J_t / K_t$. Since $\widetilde{G}_{J,t} = \kappa(j_t - \delta)$:

$$q_t = (1 + \kappa(j_t - \delta))/\mu_t \quad \Longrightarrow \quad \boxed{j_t = \delta + (\mu_t q_t - 1)/\kappa.}$$

$\hookrightarrow$ **Economic interpretation:**

- Holding expend. intensity j_t fixed, higher efficiency ($\mu_t \uparrow$) reduces the marginal cost per unit ($q_t \downarrow$).
- Holding marginal q_t fixed, higher efficiency ($\mu_t \uparrow$) raises the optimal expenditure intensity ($j_t \uparrow$).

$\trianglerighteq$ W/out adjustment costs, $q_t = 1/\mu_t = p_t^I$: marginal q equals the relative price of installed capital. The labor FOC remains unchanged, $Z_t F_2(K_t, Z_t L_t) = w_t$.

IST (2/2): Returns and Technology Risk

- Holding expenditure J_{t+1} fixed, the envelope condition remains:

$$V_{K,t+1} = F_{K,t+1} - \widetilde{G}_{K,t+1} + (1-\delta)q_{t+1}, \qquad q_t = \mathbb{E}_t[M_{t+1}V_{K,t+1}].$$

- Return on Investment. Substituting the IST investment FOC:

$$R^{I,g}_{t+1} = \frac{V_{K,t+1}}{q_t} = \frac{F_{K,t+1} - \widetilde{G}_{K,t+1} + (1-\delta)(1+\widetilde{G}_{J,t+1})/\mu_{t+1}}{(1+\widetilde{G}_{J,t})/\mu_t}.$$

- As before, consider an additional unit of capital at $t+1$. To keep same K_{t+2}, the firm reduces expenditure:

$$\mathrm{d}K_{t+2} = 0 \Longrightarrow \mathrm{d}J_{t+1} = -\frac{1-\delta}{\mu_{t+1}}\mathrm{d}K_{t+1} \iff -(1+\widetilde{G}_{J,t+1})\mathrm{d}J_{t+1} = (1-\delta)q_{t+1}\mathrm{d}K_{t+1}.$$

  ↪ the continuation value reflects the saving on future investment expenditure, adjusted for its efficiency.

- **Pricing:** Firm optimality still implies $1 = \mathbb{E}_t[M_{t+1}R^{I,g}_{t+1}]$: $M_{t+1} = \beta\frac{u_C(C_{t+1},\ell_{t+1})}{u_C(C_t,\ell_t)}$.
  IST does not change it, but affects equilibrium consumption, investment, and hence the realized SDF.

↪ **Does better IST imply a lower SDF?** Not necessarily. Higher efficiency increases investment opportunities, but exploiting them may reduce current consumption. The effect on marginal utility depends on the equilibrium.

▷ The risk premium depends on the covariance between IST-related payoffs and the SDF (i.e., assets paying in high-M states provide insurance and command lower premia).

Two Distinct Wedges: The General Formula

- *What if the installation process itself also depends on expenditure and existing capital?*
- Installation function Φ & Adjustment costs G:

$$K_{t+1} = (1-\delta)K_t + \mu_t \Phi(J_t, K_t), \qquad D_t = Y_t - w_t L_t - J_t - G(J_t, K_t).$$

- **Investment FOC:** An additional unit of expenditure costs $1 + G_{J,t}$ and installs $\mu_t \Phi_{J,t}$ units of capital:

$$-1 - G_{J,t} + q_t \mu_t \Phi_{J,t} = 0 \quad \Longrightarrow \quad q_t = (1 + G_{J,t})/\mu_t \Phi_{J,t}.$$

  ↪ Marginal q reflects both the resource cost and the efficiency of installing capital.
- Differentiating the date-$(t+1)$ value function wrt inherited capital, holding J_{t+1} fixed:

$$V_{K,t+1} = \underbrace{F_{K,t+1} - G_{K,t+1}}_{\text{net marginal output}} + \underbrace{q_{t+1}(1 - \delta + \mu_{t+1}\Phi_{K,t+1})}_{\text{continuation value}}.$$

  Unlike the baseline model, inherited capital can also affect how efficiently new capital is installed through $\Phi_{K,t+1}$.
- Combining the FOC and the capital Euler equation:

$$R^{I,g}_{t+1} = \frac{F_{K,t+1} - G_{K,t+1} + q_{t+1}(1 - \delta + \mu_{t+1}\Phi_{K,t+1})}{(1 + G_{J,t})/(\mu_t \Phi_{J,t})}, \qquad 1 = \mathbb{E}_t[M_{t+1} R^{I,g}_{t+1}].$$

↪ **Special cases:** $\Phi(J, K) = J$ recovers our previous IST model; $G = 0$ isolates installation frictions.

▷ Resource costs affect dividends; installation efficiency affects capital accumulation.

In-Class Exercise 2: Investment and Risk Premia

- **Setup:** Consider two firms, A and B, with quadratic adjustment costs, $\kappa = 4$, $\delta = 0.10$, and $\mu_t = 1$. Their marginal capital payoffs $H^i_{t+1} = V^i_{K,t+1}$ include output, adjustment-cost effects, and continuation value.
- There are two equally likely states, with the following SDF and marginal payoffs:

	State 1	State 2
SDF M_{t+1}	1.20	0.80
Firm A: H^A_{t+1}	1.00	1.60
Firm B: H^B_{t+1}	1.20	1.40

Tasks (15 minutes):

1. **Value capital and determine investment.** Compute $q_i = \mathbb{E}_t[M_{t+1}H^i_{t+1}]$ and use the investment FOC to find each optimal investment rate $x_i = J_i/K_i$.
2. **Compute expected returns.** Find the risk-free gross return $R^g_{f,t}$, each expected investment return $\mathbb{E}_t[R^{I,g}_{i,t+1}]$, and its risk premium. Verify $1 = \mathbb{E}_t[M_{t+1}R^{I,g}_{i,t+1}]$.
3. **Explain the differences.** Both firms have the same expected marginal payoff. Why do they choose different investment rates and earn different expected returns? Compute $\mathrm{Cov}_t(M_{t+1}, H^i_{t+1})$.
4. **Counterfactual:** Reverse the SDF values across states, holding marginal payoffs fixed. Recompute q_i, x_i, and risk premia. How could an IST improvement that diverts resources from consumption affect the interpretation of State 2?

Exercise 2 Solution (1/2)

1. **Step 1 - Price marginal capital.** Recall $q_i = \mathbb{E}_t[M_{t+1} H^i_{t+1}]$. With equally likely states:

$$q_A = \tfrac{1}{2}(1.20)(1.00) + \tfrac{1}{2}(0.80)(1.60) = \boxed{1.24},$$
$$q_B = \tfrac{1}{2}(1.20)(1.20) + \tfrac{1}{2}(0.80)(1.40) = \boxed{1.28}.$$

   Both firms have $\mathbb{E}_t[H^i_{t+1}] = 1.30$, but $q_B > q_A$ because firm B pays relatively more in the high-M state.

2. **Step 2 - Determine optimal investment.** Since $\mu_t = 1$, the investment FOC gives $q_i = 1 + \kappa(x_i - \delta)$:

$$x_A = 0.10 + \frac{1.24 - 1}{4} = \boxed{0.16}, \quad x_B = 0.10 + \frac{1.28 - 1}{4} = \boxed{0.17}.$$

   ↪ Higher marginal valuation encourages investment: firm B invests more.

3. **Step 3 - Compute expected returns and risk premia.** First: $\mathbb{E}_t[M_{t+1}] = 1 \Longrightarrow R^g_{f,t} = 1$. Since $R^{I,g}_{i,t+1} = H^i_{t+1}/q_i$:

	Firm A	Firm B
$\mathbb{E}_t[R^{I,g}_{i,t+1}]$	$1.30/1.24 = 1.050$	$1.30/1.28 = 1.016$
Risk premium	4.84%	1.56%
Euler check: $\mathbb{E}_t[M R^{I,g}_i]$	$1.24/1.24 = 1$	$1.28/1.28 = 1$

↪ **Same expected payoff, different returns:** Firm B's capital is more valuable ($q_B > q_A$); it invests more ($x_B > x_A$) and earns a lower expected return.

Exercise 2 Solution (2/2)

4. **Step 4 - Identify the source of the risk premia.** Recall: $\mathbb{E}_t[R_i^{I,g}] - R_f^g = -\frac{\mathrm{Cov}_t(M, H^i)}{q_i \mathbb{E}_t[M]}$.
   Since $\mathbb{E}_t[M] = 1$ and $\mathbb{E}_t[H^A] = \mathbb{E}_t[H^B] = 1.30$:

$$\mathrm{Cov}_t(M, H^A) = \mathbb{E}_t[MH^A] - 1.30 = \boxed{-0.06}, \quad \mathrm{Cov}_t(M, H^B) = \mathbb{E}_t[MH^B] - 1.30 = \boxed{-0.02}.$$

   Thus, the risk premia are $0.06/1.24 = 4.84\%$ and $0.02/1.28 = 1.56\%$. Firm A delivers relatively less in high-M State 1, when consumption is most valuable.
   ↪ It provides less insurance and therefore commands a higher risk premium.
5. **Step 5 - Reverse the SDF.** Now let $M = (0.80, 1.20)$, holding the physical payoffs unchanged. For example:

$$q_A = \tfrac{1}{2}(0.80)(1.00) + \tfrac{1}{2}(1.20)(1.60) = 1.36.$$

   Repeating the calculations for both firms:

Reversed SDF	Firm A	Firm B
Marginal value q_i	1.36	1.32
Optimal investment x_i	0.19	0.18
$\mathrm{Cov}_t(M, H^i)$	$+0.06$	$+0.02$
Expected risk premium	-4.412%	-1.515%

↪ **The ranking reverses:** Firm A now pays more in high-M State 2. Its capital provides greater insurance, has a higher value, induces more investment, and earns a lower (negative) risk premium.

▷ **Connection to IST:** A positive IST shock may encourage investment so strongly that firms divert resources away from consumption. If consumption at $t+1$ falls, marginal utility rises, and so may M_{t+1}. Thus, better investment technology does not imply necessarily a low-SDF state: what matters is how the shock affects equilibrium consumption.

Market Clearing

- Consider a representative household owning the firm, with aggregate equity supply normalized to one and bonds in zero net supply.
  - Labor supply equals labor demand: $L_t^s = L_t^d = L_t$, with leisure $\ell_t = 1 - L_t$.
  - Household budget with firm's payouts: $C_t = w_t L_t + D_t = w_t L_t + \big(Y_t - w_t L_t - J_t - \widetilde{G}(J_t, K_t)\big)$.

  $\hookrightarrow$ aggregate resource and capital constraints are:

$$\boxed{C_t + J_t + \widetilde{G}(J_t, K_t) = Y_t, \qquad K_{t+1} = (1-\delta)K_t + \mu_t J_t.}$$

- Given preferences, technology, initial capital, and the stochastic processes for (Z_t, μ_t), we seek allocations and prices such that:
  1. **Households** optimize consumption, labor supply, and portfolios; their decisions determine the SDF.
  2. **Firms** optimize labor and investment, taking wages and SDF as given; their decisions determine payouts and marginal q.
  3. **Markets clear**: goods, labor, and securities satisfy the clearing conditions, and capital accumulation holds.
  4. **Intertemporal conditions** (including transversality) rule out unsustainable allocations and price bubbles.

$\triangleright$ **General equilibrium requirement:** Household and firm decisions must be optimal *at the same prices* (w mutually consistent allocations). SDF, investment, and asset returns are determined jointly.

The Equilibrium Capital Euler Equation

- Two optimality conditions must agree:
  - **Households** determine the rate of substitution: $M_{t+1} = \beta u_C(C_{t+1}, \ell_{t+1})/u_C(C_t, \ell_t)$.
  - **Firms** equate marginal cost of investment to its discounted marginal payoff: $q_t = (1 + \widetilde{G}_{J,t})/\mu_t = \mathbb{E}_t[M_{t+1} V_{K,t+1}]$.
- **The equilibrium capital Euler equation:** Substituting the household SDF and the firm's marginal payoff gives:

$$\boxed{\frac{1 + \widetilde{G}_{J,t}}{\mu_t} = \mathbb{E}_t\left[\beta \frac{u_{C,t+1}}{u_{C,t}}\left(F_{K,t+1} - \widetilde{G}_{K,t+1} + (1-\delta)\frac{1 + \widetilde{G}_{J,t+1}}{\mu_{t+1}}\right)\right].}$$

  $\hookrightarrow$ The marginal resource cost of installing capital today equals the household's valuation of its future marginal benefits.
- Household labor supply and firm labor demand imply:

$$\underbrace{u_{\ell,t}/u_{C,t}}_{\text{household MRS}} = w_t = \underbrace{Z_t F_2(K_t, Z_t L_t)}_{\text{marginal product of labor}}.$$

  Together with i) goods clearing, ii) capital accumulation, ii) initial conditions+transversality, these equations characterize an interior competitive equilibrium.
- W/out adjustment costs or IST ($\widetilde{G} = 0$, $\mu_t = 1$):

$$\boxed{1 = \mathbb{E}_t\left[\beta \frac{u_{C,t+1}}{u_{C,t}}\left(F_{K,t+1} + 1 - \delta\right)\right].}$$

$\triangleright$ **Economic interpretation:** Equilibrium requires households' willingness to postpone consumption to match the economy's ability to transform current resources into future consumption through productive investment.

Production-Based AP: Puzzles & Extensions

- *Why does introducing production make asset-pricing puzzles more challenging?*
- **Production introduces a smoothing margin:** Households do not need to consume a predetermined resource.
  - Following a negative productivity shock, firms can reduce investment to protect current consumption.
  - Smoother consumption may reduce fluctuations in marginal utility, i.e., the risk needed to explain a large premium.
- **Adjustment costs and the equity-premium puzzle:**

  Making investment costly limits consumption smoothing and increases variation in the value of installed capital.
  - For example, habit-plus-production model helps generate a substantial equity premium, but produces excessive risk-free-rate volatility.
  - Matching risky returns is insufficient: the same equilibrium must also explain safe rates, consumption...
- *What changes with more realistic firms?*
  - **Asymmetric adj. costs:** Difficult disinvestment makes some firms particularly exposed to adverse productivity shocks.
  - **Labor and intangible capital:** Hiring, training, and organizational investment introduce additional adjustment margins.
  - **Multiple sectors and IST:** Consumption and investment technologies can experience different shocks.
  - **Financing frictions:** Debt, taxes, and financing constraints affect investment decisions.

↪ **The common mechanism:** Each extension changes the firm's marginal costs, future payoffs, or the financial claim being priced. Explaining a return premium still requires understanding its covariance with the equilibrium SDF.

≥ **The empirical challenge:** A successful production-based model must jointly explain *asset prices, investment decisions, and macroeconomic allocations*, rather than fitting return anomalies in isolation.

What Have We Learned?

- One pricing architecture, two optimization problems: Households determine the SDF; firms determine investment:

$$\underbrace{M_{t+1} = \beta \frac{u_{C,t+1}}{u_{C,t}}}_{\text{household optimality}}, \qquad \underbrace{q_t = 1 + G_{I,t} = \mathbb{E}_t[M_{t+1} V_{K,t+1}]}_{\text{firm optimality}}.$$

- Production delivers an economically motivated return:

$$\underbrace{R^{I,g}_{t+1} = \frac{F_{K,t+1} - G_{K,t+1} + (1-\delta) q_{t+1}}{q_t}}_{\text{marginal payoff relative to marginal cost}}, \qquad 1 = \mathbb{E}_t[M_{t+1} R^{I,g}_{t+1}].$$

  ↪ Expected returns depend on profitability, investment opportunities, and the covariance of marginal payoffs with the SDF.

- Under homogeneity, $q_t = Q_t$. IST changes the marginal cost of installed capital and equilibrium allocations.

- **From theory to empirical tests:**
  1. Specify technology, capital measurement, depreciation, and adjustment costs.
  2. Construct marginal investment returns and identify the financial claims they should match.
  3. Confront expected returns, volatility, comovement, and conditional pricing restrictions jointly.

  ↪ Firm characteristics can help predict returns.

▷ **Looking ahead:** Our analysis relies on firms' ability to adjust investment and households' ability to trade at equilibrium prices. What happens when *information frictions, trading costs, and financing constraints* restrict these decisions?

Thanks for your attention! See you next time!

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